Table 3: Models, datasets, and prompting strategies used in our experiments. Models marked with † are run with a 4k context size window. Note that Gemma has a larger than 4k context size window, but VLLM only supports up to a 4k context size window for it. Models marked with * indicate closed-source models that cannot handle prefixed assistant messages. Datasets marked with △ do not have a few-shot setting.

Models	Llama 2 7B Chat † (Touvron et al., 2023), Mistral 7B Instruct v0.3 (Jiang et al., 2023), Llama 3.1 8B Instruct (Dubey et al., 2024), Llama 3.1 70B Instruct, Gemma 2 9B It† (Riviere & et. al, 2024), Phi-3 Small 8k Instruct (Abdin et al., 2024), gpt-4o-mini-2024-07-18*, gpt-4o-2024-08-06*, Gemini 1.5 Flash*(Reid & et. al, 2024), Gemini 1.5 Pro*(Reid & et. al, 2024), claude-3-haiku-20240307* (Anthropic, a), claude-3-5-sonnet-20240620* (Anthropic, b)
Datasets	CommonsenseQA (Talmor et al., 2019), StrategyQA (Geva et al., 2021), SiQA△ Sap et al. (2019), PiQA△ (Bisk et al., 2019), Winogrande△ (Sakaguchi et al., 2021), GPQA (Rein et al., 2023), MuSR (Sprague et al., 2024), ContextHub (Levels 1 and 2 only) (Hua et al., 2024), ARC△ (Clark et al., 2018), AGIEval LSAT (Zhong et al., 2023), MMLU (Hendrycks et al., 2021a), MMLU Pro (Wang et al., 2024), MATH (Hendrycks et al., 2021b), GSM8K (Cobbe et al., 2021), GSM8K-hard (Gao et al., 2023), FOLIO (Han et al., 2022), MuSiQue△ (Trivedi et al., 2022), Big-Bench Hard (Suzgun et al., 2023; Srivastava et al., 2022), BiGGen Bench (Kim et al., 2024)
Prompts	zero-shot direct answer, zero-shot CoT (Kojima et al., 2022), few-shot direct answer (Brown et al., 2020), few-shot CoT (Wei et al., 2022)

Table 4: List of datasets used in our experiments. We categorize each dataset into one of five categories based on the type of reasoning required: Commonsense, Knowledge, Soft Reasoning, Symbolic, or Mathematical. We also report answer formats. When we use few-shot prompts, we mark how many examples those prompts contain. BiGGen Bench has many categories of questions that explicitly ask for CoTs in the response; we ignore those categories for our evaluation.

Dataset	Type	Answer Format	m-Shots
CommonsenseQA	Commonsense	Multiple choice	7
StrategyQA	Commonsense	True or False	6
SIQA	Commonsense	Multiple choice	0
PIQA	Commonsense	Multiple choice	0
Winogrande	Commonsense	Multiple choice	0
Arc Easy	Knowledge	Multiple choice	0
Arc Challenge	Knowledge	Multiple choice	0
AGIEval LSAT	Soft Reasoning	Multiple choice	3
BiGGen-Bench	Soft Reasoning	Free response	0
MMLU	Knowledge	Multiple Choice	5
MMLU Pro	Knowledge	Multiple Choice	5
BigBench-Hard	Symbolic	Multiple Choice	0
MuSR	Soft Reasoning	Multiple Choice	1
GPQA	Mathematical	Multiple Choice	3
MuSiQue	Soft Reasoning	Short Answer	0
GSM8K	Mathematical	Short Answer	8
GSM8K-Hard	Mathematical	Short Answer	8
FOLIO	Symbolic	True, False, or Unknown	4
ContextHub	Symbolic	True, False, or Neither	3
MATH	Mathematical	Short Answer	4

Table 5: List of models for our experiments. We focus on contemporary instruction-tuned models; although pretrained and smaller language models could be used, they are not the focus of our study. Prompts and outputs used for each model are available on Huggingface. * Note that Gemma can accept more than 4k input tokens, but we are restricted to 4k by vLLM.

Model	Context Length	Is Open Source
Llama 2 7B Chat	4k	True
Mistral 7B Instruct v0.3	8k	True
Llama 3.1 8B Instruct	128k	True
Llama 3.1 70B Instruct	128k	True
Gemma 2 9B It	4k*	True
Qwen 7B Instruct	131k	True
Qwen 72B Instruct	131k	True
GPT4o-Mini	128k	False
GPT4o	128k	False
Gemini 1.5 Pro	128k	False
Gemini Flash	1m	False
Claude 3.5 Sonnet	200k	False
Claude 3 Haiku	200k	False

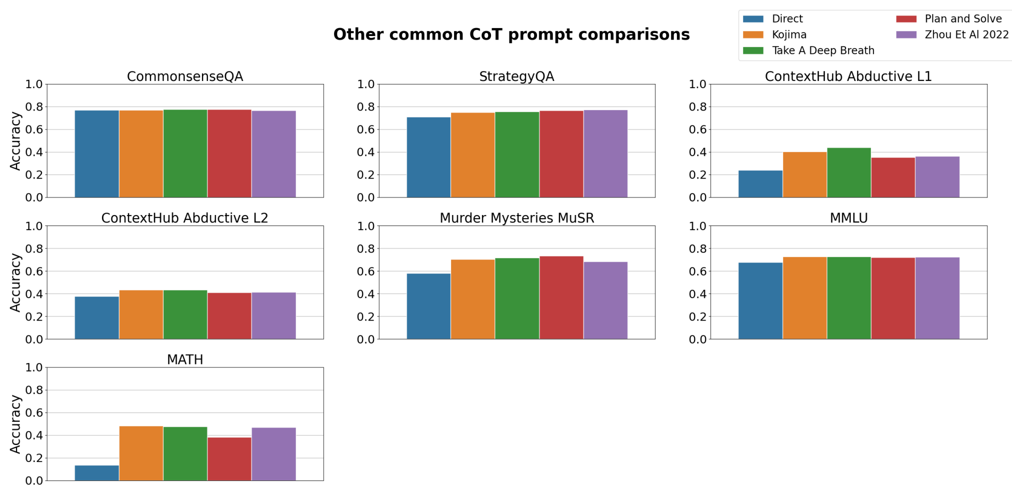

Figure 7: Performance of multiple prompts commonly used to elicit reasoning through CoT in the zero shot setting. Each prompt starts the assistant completion with a different phrase meant to elicit reasoning. All results are from using Llama 3.1 8B Instruct. For the Kojima variant, we explicitly place "*Let's think step by step.*" in the assistant message. There is very little variation between the CoT prompts on average.

Figure 7 shows variation due to prompts is typically small and no prompt gives a consistent gain over the other. For our experiments, this suggests that different prompts have small effects on the overall outcome on average.

E Few-shot Experiments

Compared to a zero-shot prompt, a few-shot prompt additionally contains demonstrations of the relevant reasoning mode on different problem instances $\{(v(\mathbf{q}_i), \mathbf{y}_i^*)\}$. Few-shot prompts for direct answer simply encode the answer a_i as $\mathbf{y}_i^*$, whereas few-shot prompts for chain-of-thought include a reasoning trace ending in the correct answer. Now we can define the m-shot direct prompt as

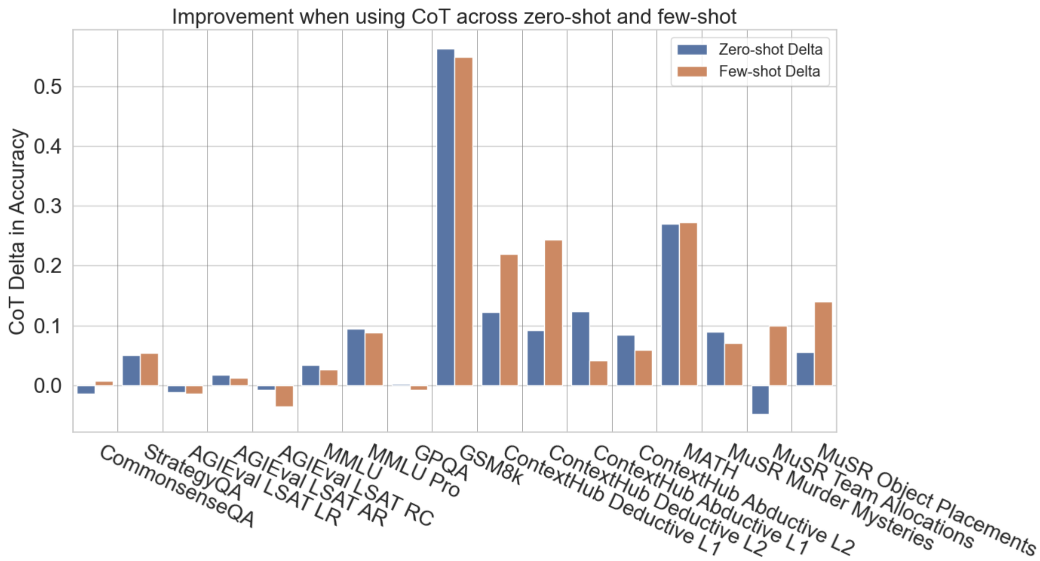

Figure 8: Average performance improvement from using CoT across different models in the zero-shot and few-shot settings. Each bar represents how much CoT improves the accuracy for that specific setting. In general, CoT in the few-shot setting does not change the qualitative performance of CoT versus zero-shot, though it can change the magnitude for symbolic datasets.

$\mathcal{I}_{\text{da}}^m(\mathbf{q}) = v_{\text{da}}(\mathbf{q}_1)\mathbf{a}_1 v_{\text{da}}(\mathbf{q}_2)\mathbf{a}_2 \ldots v_{\text{da}}(\mathbf{q}_m)\mathbf{a}_m v_{\text{da}}(\mathbf{q})$ and the m-shot cot prompt as $\mathcal{I}_{\text{cot}}^m(\mathbf{q}) = v_{\text{cot}}(\mathbf{q}_1)\mathbf{y}_1^* v_{\text{cot}}(\mathbf{q}_2)\mathbf{y}_2^* \ldots v_{\text{cot}}(\mathbf{q}_m)\mathbf{y}_m^* v_{\text{cot}}(\mathbf{q})$.

Figure 8 shows the difference between few-shot prompting and the zero-shot setting discussed in the main text of the paper. We see that using CoT in the few-shot setting largely does not change the datasets that benefit from it. Only one dataset, MuSR Team Allocation, starts to improve with few-shot; however, we believe this to be an exception because the final step to derive the answer is complex in the prompt and clearer in the examples. The magnitude of improvement over direct answer prompting when using CoT is also similar to the zero-shot setting.

F Expanded CoT vs Direct Experimental Results

F.1 Full Zero-shot Results